Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The magnetic susceptibility of a material of a rod is $499.$ Permeability of vaccum is $4\pi \times 10^{-7}$ henry/m. Absolute permeability of the material of the rod in henry per meter is

(A) $\pi \times 10^{-4}$

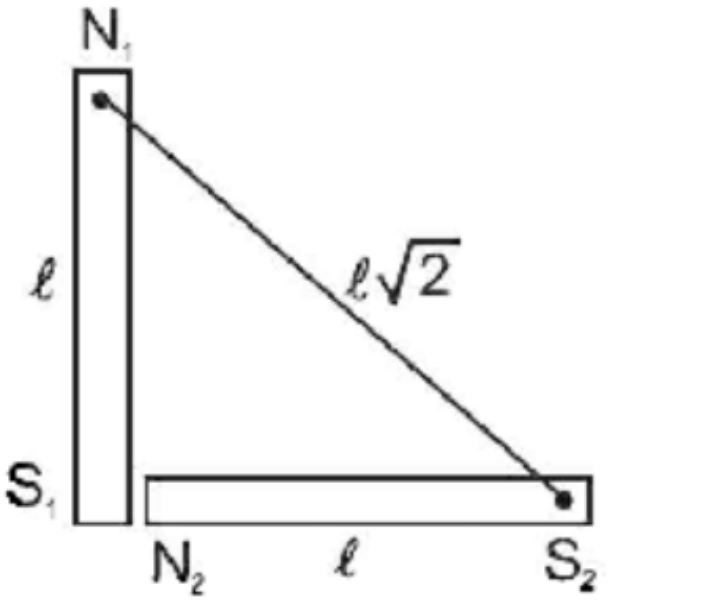
(B) $2\pi \times 10^{-4}$

(C) $3\pi \times 10^{-4}$

(D) $4\pi \times 10^{-4}$

Q.2

Two identical thin bar magnets each of length ℓ and pole strength m are placed at right angles to each other, with north pole of one touching south pole of the other, then the magnetic moment of the system is



(A) $1\ \mathrm{m}\ell$

(B) $2\ \mathrm{m}\ell$

(C) $\sqrt{2}\ \mathrm{m}\ell$

(D) $\mathrm{m}\ell/2$

Q.3

A coil in the shape of an equilateral triangle of side l is suspended between the pole pieces of a permanent magnet such that $\vec{B}$ is in the plane of the coil. If due to a current i in the triangle a torque τ acts on it, the side l of the triangle is

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(A) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{\mathrm{Bi}}\right)^{\frac{1}{2}}$

(B) $2\left(\frac{\tau}{\sqrt{3}\,\mathrm{B{\cdot}i}}\right)^{\frac{1}{2}}$

(C) $\frac{2}{\sqrt{3}}\left(\frac{\tau}{\mathrm{B.i}}\right)$

(D) $\frac{1}{\sqrt{3}}\frac{\tau}{\mathrm{Bi}}$

Q.4

The length of a magnet is large compared to its width and breadth. The time period of its oscillation in a vibration magnetometer is $2\ \mathrm{s}$. The magnet is cut along its length into three equal parts and these parts are then placed on each other with their like poles together. The time period of this combination will be

(A) $2\sqrt{3}\ \mathrm{s}$

(B) $\frac{2}{3}\ \mathrm{s}$

(C) $2\ \mathrm{s}$

(D) $\frac{2}{\sqrt{3}}\ \mathrm{s}$

Q.5

A magnetic dipole is under the influence of two magnetic fields The angle between the field directions is 60° and one of the fields has a magnitude of $1.2 \times 10^{-2}\ \mathrm{T}$. If the dipole comes to stable equilibrium at an angle of 15° with this field, what is the magnitude of other field?

(A) 4.4×10^{-3} tesla

(B) 5.2×10^{-3} tesla

(C) 3.4×10^{-3} tesla

(D) 7.8×10^{-3} tesla

Q.6

A bar magnet $8\ \mathrm{cm}$ long is placed in the magnetic merdian with the N -pole pointing towards geographical north. Two neutral points separated by a distance of $6\ \mathrm{cm}$ are obtained on the equatorial axis of the magnet. If horizontal component of earth's field $= 3.2 \times 10^{-5}\ \mathrm{T}$, then pole strength of magnet is

(A) $5\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

(B) $10\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

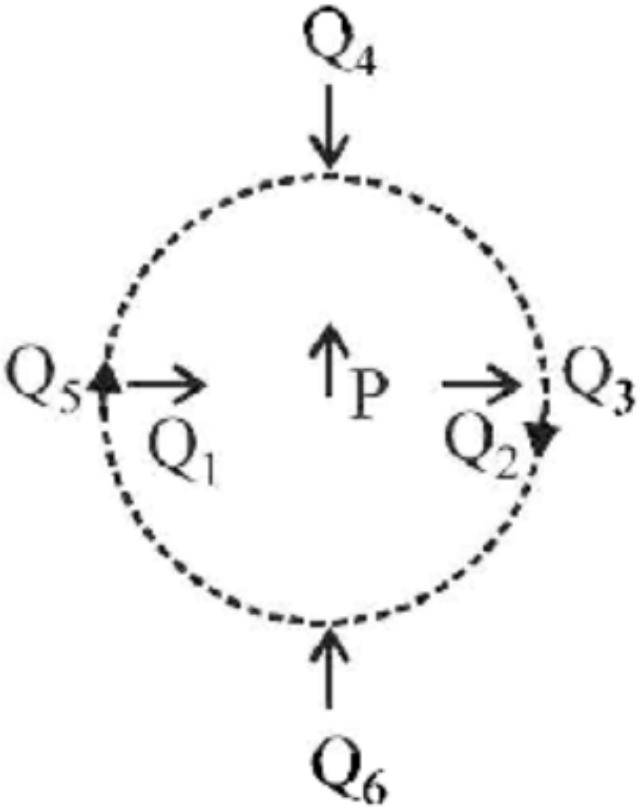
(C) $2.5\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

(D) $20\mathrm{ab} - \mathrm{amp} \times \mathrm{cm}$

Q.7

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The figure shows the various positions (labelled by subscripts) of small magnetised needles $\mathbf{P}$ and $\mathbf{Q}$. The arrows show the direction of their magnetic moment. Which configuration corresponds to the lowest potential energy among all the configurations shown?



(A) PQ_3

(B) PQ_4

(C) PQ_5

(D) PQ_6

Q.8

A vibration magnetometer consists of two identical bar magnets placed one over the other such that they are perpendicular and bisect each other. The time period of oscillation in a horizontal magnetic field is $2^{5/4}$ seconds. One of the magnets is removed and if the other magnet oscillates in the same field, then the time period in seconds is

(A) $2^{1/4}$

(B) $2^{1/2}$

(C) 2

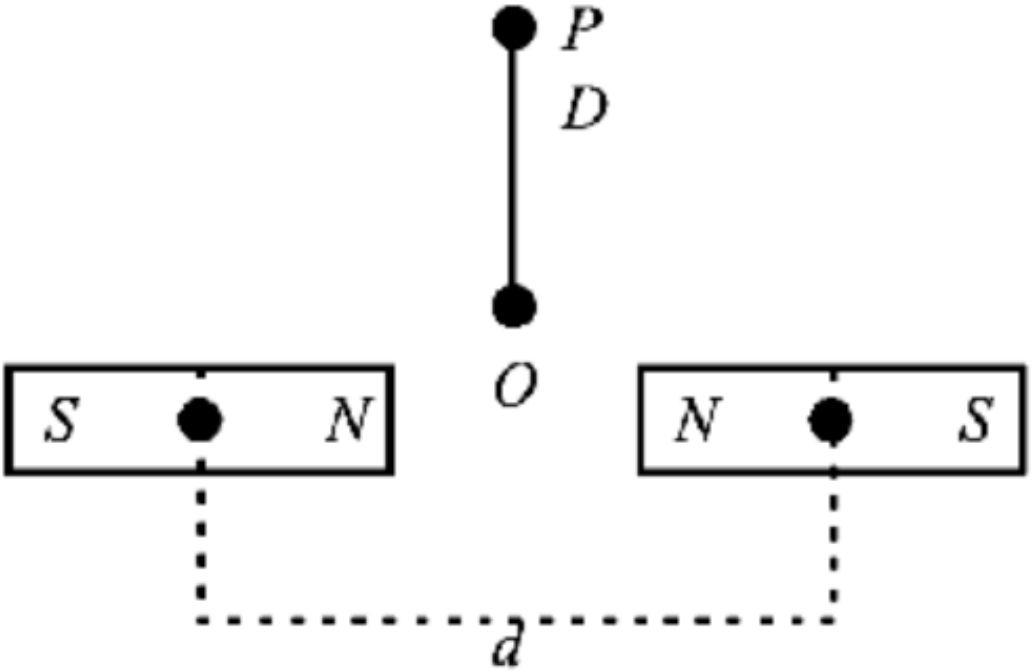
(D) $2^{3/4}$

Q.9

Two identical bar magnets are fixed with their centres at a distance d apart. A stationary charge Q is placed at

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P in between the gap of the two magnets at a distance D from the centre O as shown in the Figure



The force on the charge Q is

(A) directed perpendicular to the plane of paper

(B) zero

(C) directed along OP

(D) directed along PO

Q.10

Two short bar magnets P and Q are arranged such that their centres are on the X-axis and are separated by a large distance. The magnetic axes of P and Q are along X and Y axes respectively. At a point R, midway between their centres, if B is the magnitude of induction due to Q, the magnitude of total induction at R due to the both magnets is

(A) $3\ \mathrm{B}$

(B) $\sqrt{5}\ \mathrm{B}$

(C) $\frac{\sqrt{5}}{2}\ \mathrm{B}$

(D) B

Q.11

A thin rectangular magnet suspended freely has a period of oscillation equal to T. Now it is broken into two equal halves (each having half of the original length) and one piece is made to oscillate freely in the same field. If its period of
oscillation is T', the ratio $\frac{T'}{T}$ is

(A) $\frac{1}{2\sqrt{2}}$

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(B) $\frac{1}{2}$

(C) 2

(D) $\frac{1}{4}$

Q.12

Two identical magnetic dipoles of magnetic moments $1.0\ \mathrm{A}-\mathrm{m}^2$ each, placed at a separation of $2\ \mathrm{m}$ with their axis perpendicular to each other. Then what is the resultant magnetic field in tesla at point midway between the dipole? (Enter the number which is power of 10 in the answer expression)

(A) -5

(B) -6

(C) -8

(D) -7

Q.13

A bar magnet of moment of inertia $9 \times 10^{-5}\ \mathrm{kg\ m}^2$ placed in a vibration magnetometer and oscillating in a uniform magnetic field $16\pi^2 \times 10^{-5}\ \mathrm{T}$ makes 20 oscillations in $15\ \mathrm{s}$. Then what will be the value of magnetic moment of the bar magnet?

(A) 1

(B) 2

(C) 3

(D) 4

Q.14

Two short bar magnets of length $1\ \mathrm{cm}$ each have magnetic moments $1.20\mathrm{Am}^2$ and $1.00\mathrm{Am}^2$ respectively. They are placed on a horizontal table parallel to each other with their N poles pointing towards the South. They have a common magnetic equator and are separated by a distance of $20.0\ \mathrm{cm}$. Then find the close value of the resultant horizontal magnetic induction (in $\mathrm{wb/m}^2$) at the mid-point O of the line joining their centres? (Horizontal component of earth's magnetic induction is $3.6 \times 10^{-5}\ \mathrm{Wb/m}^2$)

(A) 0.002056

(B) 0.000265

(C) 0.002065

(D) 0.000256

Q.15

The coercivity of a small magnet where the ferromagnet gets demagnetized is $3 \times 10^3\ \mathrm{Am}^{-1}$. Then what is

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the value of current (in A) required to be passed in a solenoid of length $10\ \mathrm{cm}$ and number of turns 100 , so that the magnet gets demagnetized when inside the solenoid?

(A) 2

(B) 4

(C) 5

(D) 3

Q.16

If a bar magnet of pole strength m and magnetic moment M is cut perpendicular to its axis in two equal halves then its new pole strength m' and magnetic moment M' are respectively

(A) $\mathrm{m}' = \mathrm{m}$ and $\mathrm{M}' = \mathrm{M}$

(B) $\mathrm{m}' = \mathrm{m}$ and $\mathrm{M}' = \frac{\mathrm{M}}{2}$

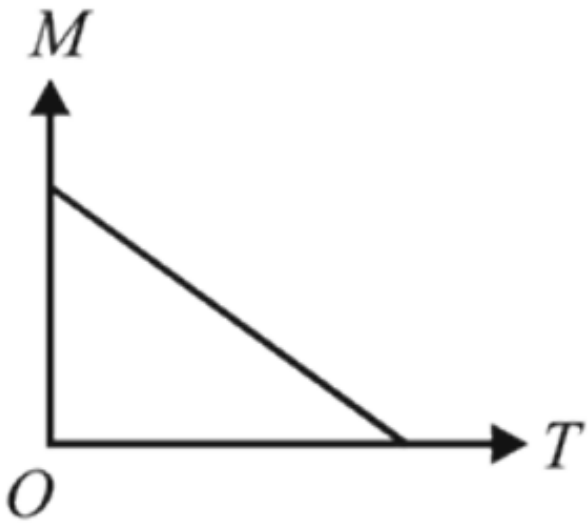
(C) $\mathrm{m}' = \frac{\mathrm{m}}{2}$ and $\mathrm{M}' = 2\mathrm{M}$

(D) $\mathrm{m}' = 2\ \mathrm{m}$ and $\mathrm{M}' = \frac{\mathrm{M}}{2}$

Q.17

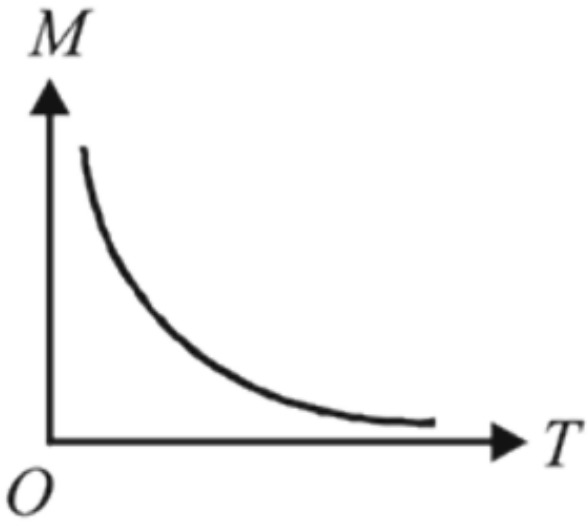
A curve between magnetic moment and temperature of magnet is

(A)

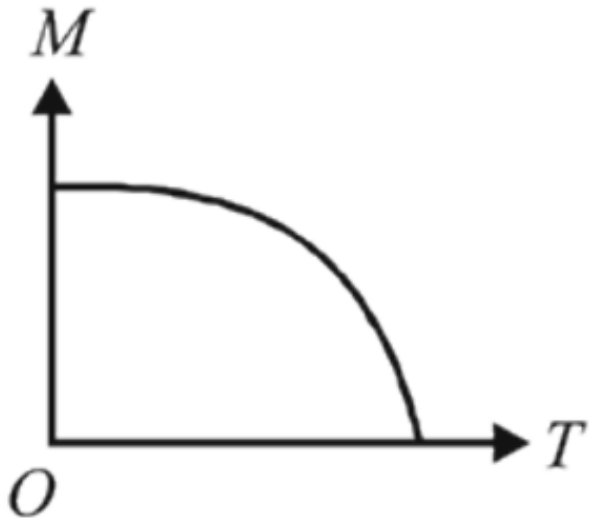


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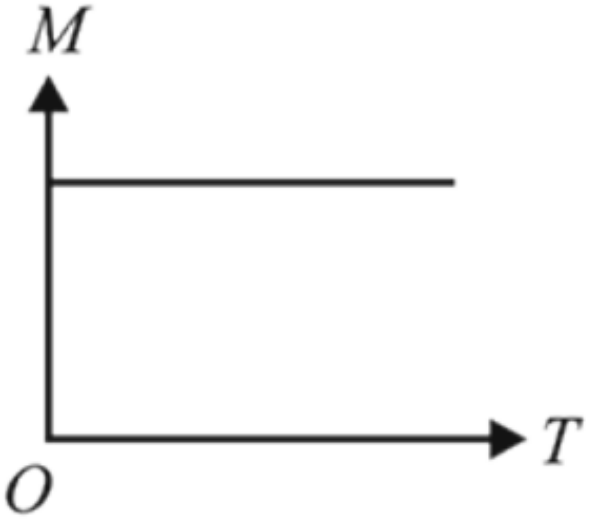
(B)



(C)



(D)



Q.18

A bar magnet has a length $8\ \text{cm}$. The magnetic field at a point at a distance $3\ \text{cm}$ from the centre in the broad side-on position is found to be $4 \times 10^{-6} T$. The pole strength of the magnet is.

(A) $6 \times 10^{-5}\text{Am}$

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(B) $5 \times 10^{-5} \mathrm{Am}$

(C) $2 \times 10^{-4} \mathrm{Am}$

(D) $3 \times 10^{-4} \mathrm{Am}$

Q.19

A permanent magnet in the shape of a thin cylinder of length $10 \mathrm{~cm}$ has magnetisation $(M)=10^{6} \mathrm{~A} \mathrm{~m}^{-1}$. Its magnetization current I_{M} is

(A) $10^{5} \mathrm{~A}$

(B) $10^{6} \mathrm{~A}$

(C) $10^{7} \mathrm{~A}$

(D) $10^{8} \mathrm{~A}$

Q.20

Assume that each iron atom has a permanent magnetic moment equal to 2 Bohr magnetons $(1$ Bohr magneton $=9.27 \times 10^{-24} \mathrm{~A}-\mathrm{m}^{2})$. The density of atoms in iron is 8.52×10^{28} atoms $/ \mathrm{m}^{3}$. Find the maximum magnetic moment per unit volume.

(A) $1.5 \times 10^{2} \mathrm{~A} \mid \mathrm{m}$

(B) $1.58 \times 10^{6} \mathrm{~A} \mid \mathrm{m}$

(C) $1.2 \times 10^{5} \mathrm{~A} \mid \mathrm{m}$

(D) $1.3 \times 10^{6} \mathrm{~A} \mid \mathrm{m}$

Answers & Solutions

Q.1 Answer:

$2\pi \times 10^{-4}$

Solution:

$\mu=\mu_{0}(1+\chi); \mu=500 \times 4\pi \times 10^{-7}=2\pi \times 10^{-4}$ **henry/metre**

$\therefore \quad (b)$

Q.2 Answer:

$\sqrt{2}\, \mathrm{m}\ell$

Solution:

Initial magnetic moment of each magnet $=\mathrm{m} \times \ell$**. As is clear from Fig.,** S_{1} **and** N_{2} **neutralize each other. Effective distance between,**

N_{1} **and** $\mathrm{S}_{2}=\sqrt{\ell^{2}+\ell^{2}}=\ell\sqrt{2}$

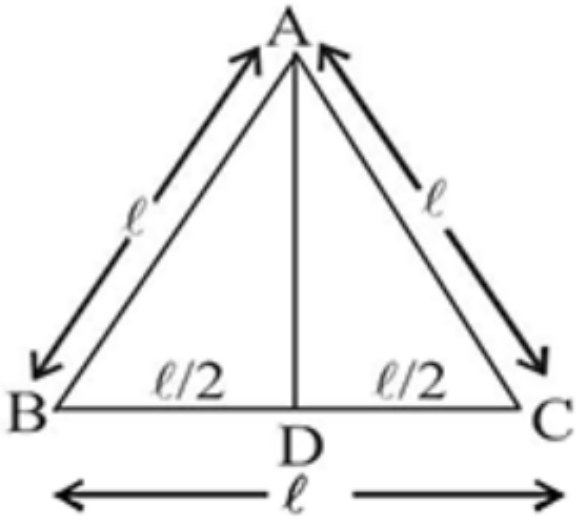
$\therefore \mathrm{M}'=\mathrm{m}\ell\sqrt{2}$

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Q.3 Answer:

$2\left(\frac{\tau}{\sqrt{3}\,\mathrm{B}\cdot\mathrm{i}}\right)^{\frac{1}{2}}$

Solution:



$\tau = \mathrm{MB}\sin\theta$

$\tau = \mathrm{iAB}\sin 90^\circ$

$\therefore \mathrm{A} = \frac{\tau}{\mathrm{iB}}$

(AD) Also, $\mathrm{A} = 1/2(\mathrm{BC})$

But $\frac{1}{2}(\mathrm{BC})(\mathrm{AD}) = \frac{1}{2}(l)\sqrt{l^2 - \left(\frac{l}{2}\right)^2} = \frac{\sqrt{3}}{4}l^2$

$\Rightarrow \quad \frac{\sqrt{3}}{4}(l)^2 = \frac{\tau}{\mathrm{Bi}}$

$\therefore$

$l = 2\left(\frac{\tau}{\sqrt{3}\,\mathrm{B.i}}\right)^{\frac{1}{2}}$

Q.4 Answer:

$\frac{2}{3}$ s

Solution:

$T = 2\pi\sqrt{\frac{I}{M\times B}} = 2\pi\sqrt{\frac{I}{MB}}$ **where** $I = \frac{1}{12}m\ell^2$

When the magnet is cut into three pieces the pole strength will remain the same and

M.I. $(I') = \frac{1}{12}\left(\frac{m}{3}\right)\left(\frac{\ell}{3}\right)^2 \times 3 = \frac{I}{9}$

We have, Magnetic moment (M) **= Pole strength** $(m) \times \ell$ **. New magnetic moment,** $M' = m \times \left(\frac{\ell}{3}\right) \times 3 = m\ell = M$

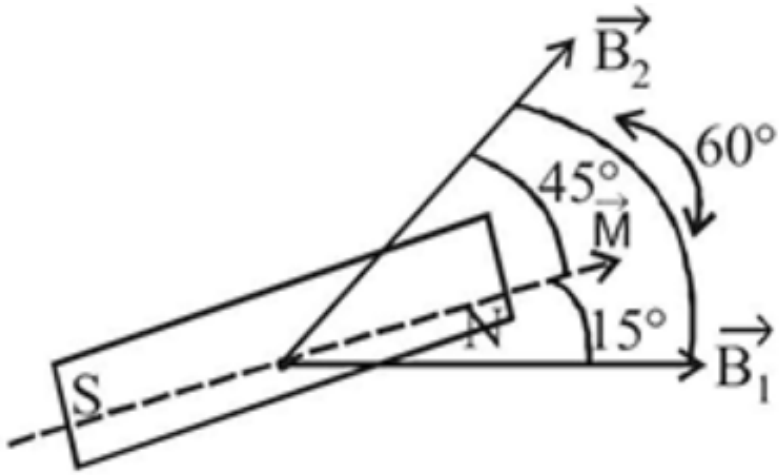
$\therefore T' = \frac{T}{\sqrt{9}} = \frac{2}{3}s.$

Q.5 Answer:

4.4×10^{-3} tesla

Solution:

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Given that : $B_1 = 1.2 \times 10^{-2}$ T, **orientation of dipole with the field** $B_1, \theta_1 = 15^\circ$

Hence, orientation of dipole with $B_2, \theta_2 = 60^\circ - 15^\circ = 45^\circ$ **(figure)**

As the dipole is in equilibrium, therefore, the torque on the dipole due to the two fields must be equal and opposite. If M **be the magnetic dipole moment of the dipole, then** $\tau_1 = \tau_2$ **or**

$$MB_1 \sin\theta_1 = MB_2 \sin\theta_2$$

or, $B_2 = \frac{B_1 \sin\theta_1}{\sin\theta_2} = \frac{1.2\times10^{-2}\sin 15^\circ}{\sin 45^\circ}$

$= \frac{1.2\times10^{-2}\times0.2588}{0.7071} = 4.4 \times 10^{-3}$ **Tesla**

Q.6 Answer:

$5\text{ab} - \text{amp} \times \text{cm}$

Solution:

Here, $2\ell = 8$ cm, $\ell = 4$ cm, $\text{d} = \frac{6}{2} = 3$ cm.

At neutral point,

$$\text{H} = \text{B} = \frac{\mu_0}{4\pi}\frac{\text{M}}{(\text{d}^2 + \ell^2)^{3/2}}$$

$$= 10^{-7}\frac{\text{M}}{(5\times10^{-2})^3} = \frac{\text{M}}{1250}$$

$$\therefore \text{M} = 1250\text{H} = 1250 \times 3.2 \times 10^{-5}\text{Am}^2$$

$$\text{m} = \frac{\text{M}}{2\ell} = \frac{1250 \times 3.2 \times 10^{-5}}{8 \times 10^{-2}}\ \text{A m}$$

$$= 0.5\text{Am} = 0.5 \times \frac{1}{10}\text{abamp} \times 100\ \text{cm}$$

$$= 5\text{ab} - \text{ampcm}$$

Q.7 Answer:

PQ_6

Solution:

PQ_6 **corresponds to the lowest potential energy among all the configurations shown.**

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Q.8 Answer:

2

Solution:

Initially magnetic moment of system $M_1 = \sqrt{M^2 + M^2} = \sqrt{2M}$ **and moment of inertia** $\mathrm{I_1} = \mathrm{I} + \mathrm{I} = 2\mathrm{I}$

Finally when one of the magnet is removed then $\mathrm{M_2} = \mathrm{M}$ **and** $\mathrm{I_2} = \mathrm{I}$

So, $T = 2\pi\sqrt{\frac{I}{MB_H}}$

$\frac{T_1}{T_2} = \sqrt{\frac{I_1}{I_2} \times \frac{M_2}{M_1}} = \sqrt{\frac{2I}{I} \times \frac{M}{\sqrt{2}M}}$

$\Rightarrow T_2 = \frac{2^{5/4}}{2^{1/4}} = 2\text{sec}$

Q.9 Answer:

zero

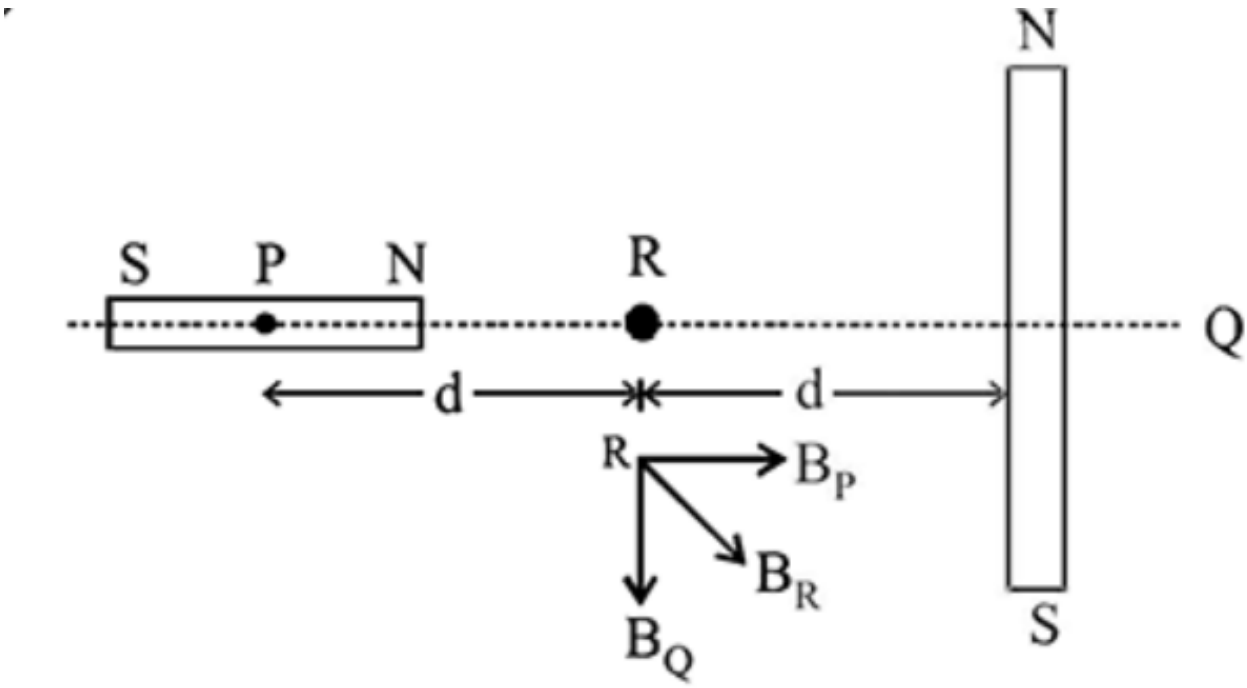
Solution:

Force on a charged particle is given by $F = qvB$**. Here** $v = 0$ **and also resultant** B **is zero.** $\therefore$ **Force** $= 0$

Q.10 Answer:

$\sqrt{5}\,\mathrm{B}$

Solution:



$$\mathrm{B_R} = \sqrt{\mathrm{B_P^2} + \mathrm{B_Q^2}} = \sqrt{\left(\frac{\mu_0}{4\pi}\frac{2\mathrm{M}}{\mathrm{d^3}}\right)^2 + \left(\frac{\mu_0}{4\pi}\frac{\mathrm{M}}{\mathrm{d^3}}\right)^2}$$

Q.11 Answer:

$\frac{1}{2}$

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Solution:

The time period of a rectangular magnet oscillating in earth's magnetic field is given by $T = 2\pi\sqrt{\frac{I}{\mu B_H}}$

where I = Moment of inertia of the rectangular magnet μ = Magnetic moment B_H = Horizontal component of the earth's magnetic field Case 1

$T = 2\pi\sqrt{\frac{I}{\mu B_H}}$ where $\mathrm{I} = \frac{1}{12}M\ell^2$

Case 2 Magnet is cut into two identical pieces such that each piece has half the original length. Then

$T' = 2\pi\sqrt{\frac{I'}{\mu' B_H}}$

where $I' = \frac{1}{12}\left(\frac{M}{2}\right)\left(\frac{\ell}{2}\right)^2 = \frac{I}{8}$ and $\mu' = \frac{\mu}{2}$

$\therefore \frac{T'}{T} = \sqrt{\frac{I'}{\mu'} \times \frac{\mu}{I}}$

$= \sqrt{\frac{I/8}{\mu/2} \times \frac{\mu}{I}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$

Q.12 Answer:

-7

Solution:

As the axes are perpendicular, mid point lies on axial line of one magnet and on equatorial line of other magnet.

$\therefore B_1 = \frac{\mu_0}{4\pi}\frac{2M}{d^3} = \frac{10^{-7}\times 2\times 1}{1^3} = 2\times 10^{-7}$

and $\mathrm{B_2} = \frac{\mu_0}{4\pi}\frac{\mathrm{M}}{\mathrm{d}^3} = 10^{-7}$

$\therefore$ Resultant field $= \sqrt{\mathrm{B_1^2 + B_2^2}} = \sqrt{5}\times 10^{-7}\ \mathrm{T}$

Q.13 Answer:

4

Solution:

Given, $I = 9\times 10^{-5}\ \mathrm{kg\ m^2}, B = 16\pi^2\times 10^{-5}\ \mathrm{T}$

$T = \frac{15}{20} = \frac{3}{4}\ \mathrm{s}$

In a vibration magnetometer Time period, $T = 2\pi\sqrt{\frac{I}{MB}}$ or $M = \frac{4\pi^2 I}{BT^2}$

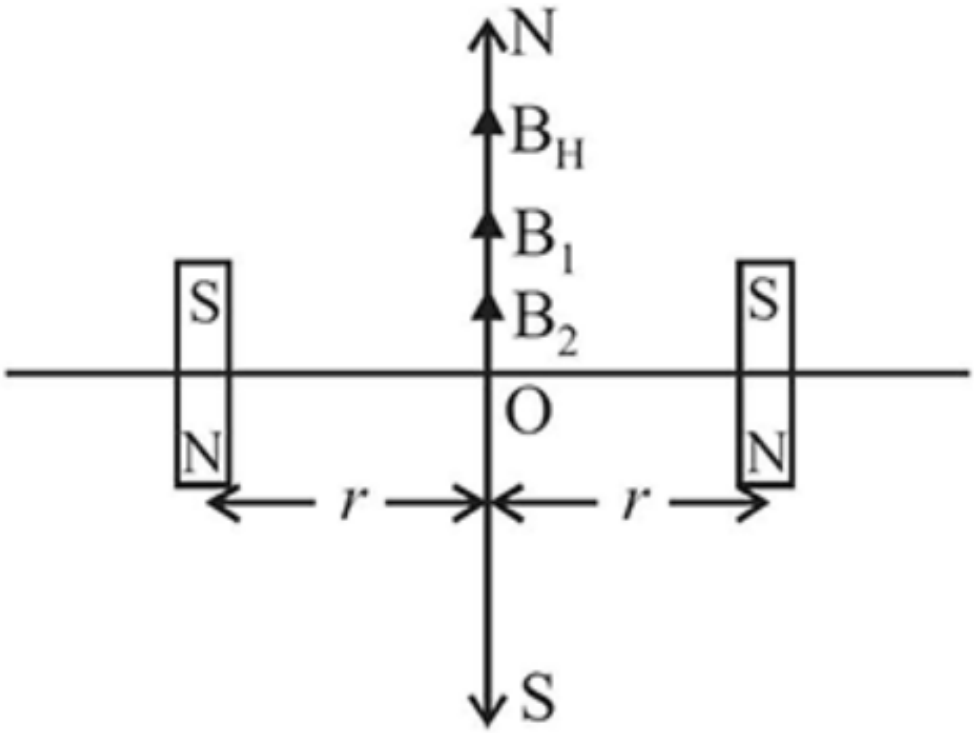
$M = \frac{4\pi^2\times 9\times 10^{-5}}{16\pi^2\times 10^{-5}\times\left(\frac{3}{4}\right)^2} = 4\ \mathrm{A\ m^2}$

Q.14 Answer:

0.000256

Solution:

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Given : $M_1 = 1.20 \mathrm{Am}^2$

$M_2 = 1.00 \mathrm{Am}^2$

$$r = \frac{20}{2} \text{ cm} = 0.1 \text{ m}$$

$$\mathrm{B_{net}} = \mathrm{B_1} + \mathrm{B_2} + \mathrm{B_H}$$

$$B_{\text{net}} = \frac{\mu_0}{4\pi} \frac{(M_1 + M_2)}{r^3} + B_H$$

$$= \frac{10^{-7}(1.2 + 1)}{(0.1)^3} + 3.6 \times 10^{-5}$$

$$= 2.56 \times 10^{-4} \text{ Wb/m}^2$$

Q.15 Answer:

3

Solution:

Magnetic field in solenoid $B = \mu_0 n \mathrm{i} \Rightarrow \quad \frac{B}{\mu_0} = ni$

(Where n **= number of turns per unit length)** $\Rightarrow \quad \frac{B}{\mu_0} = \frac{Ni}{L}$

$\Rightarrow \quad 3 \times 10^3 = \frac{100i}{10 \times 10^{-2}}$

$\Rightarrow \quad i = 3 \text{ A}$

Q.16 Answer:

$\mathrm{m}' = \mathrm{m}$ and $\mathrm{M}' = \frac{\mathrm{M}}{2}$

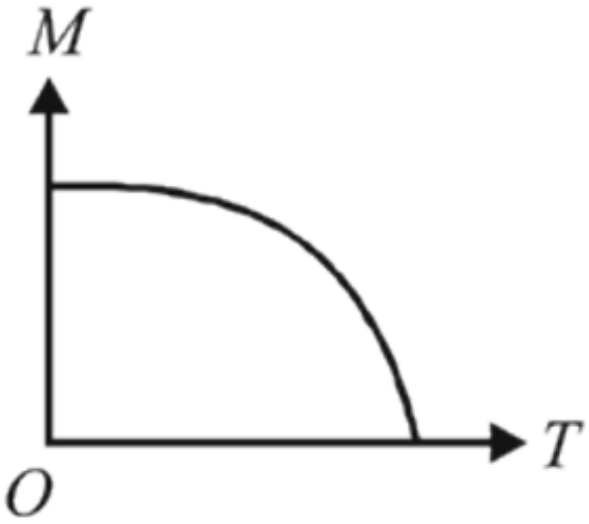
Solution:

When a bar magnet cut perpendicular to its axis into two equal parts then $\mathrm{m}' = \mathrm{m}$ **and length** $\ell' = \frac{\ell}{2}$

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$\therefore \mathrm{M}' = \mathrm{m}\ell' = \mathrm{m}\frac{\ell}{2} = \frac{\mathrm{M}}{2}$

Q.17 Answer:



Solution:

Magnetism of a magnet falls with rise of temperatue and becomes practically zero above curie temperature

Q.18 Answer:

$6 \times 10^{-5} \mathrm{Am}$

Solution:

Magnetic field due to a bar magnet in the broad-side on position is given by

$$B = \frac{\mu_0}{4\pi} \frac{M}{\left[r^2 + \frac{\ell^2}{4}\right]^{3/2}}; M = m\ell$$

After substituting the values and simplifying we get

$$B = 6 \times 10^{-5}\ \mathrm{A} - \mathrm{m}$$

Q.19 Answer:

$10^5\ \mathrm{A}$

Solution:

$I_M = ML = 10^6 \times 0.1 = 10^{-5}\ \mathrm{A}$

Q.20 Answer:

$1.58 \times 10^6\ \mathrm{A} \mid \mathrm{m}$

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Solution:

The total magnetic moment per unit volume.

$$= \frac{\left(8.52 \times 10^{28}\right) \times \left(2 \times 9.27 \times 10^{-24}\right)}{1}$$

$$= 1.58 \times 10^{6}\ \mathrm{A/m}$$

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